Samuel Mendes dos Santos

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Summary

Data Engineering Intern and Computer Science student focused on code-first pipelines, data quality, and automation with Python and SQL.

Education

Estácio University – Parangaba Campus — Fortaleza, Brazil
B.Sc. in Computer Science — *In progress*

Experience

Unitech – Unimed — Fortaleza, Brazil
Data Engineering Intern — *Aug 2026 – Present*

- Built a Pentaho-to-Python translator: parsed the PDI repository stored in the database (job, step, and connection tables) and converted each workflow into a declarative YAML file, executed by a Python engine integrated with the company's internal library for Oracle operations. As a result, workflows no longer depend on the legacy PDI and are now maintained in Python, making them easier to maintain.
- Developed a monthly API extraction in Python with date-range queries (start to end of month), using batching and `executemany` for bulk database inserts, which reduced the process runtime.
- Automated data collection from REST APIs and web pages using Python, Pandas, and Playwright (web scraping), feeding the results into the team's data routines.
- Write advanced SQL queries (joins, window functions, subqueries) and apply data quality checks to healthcare insurance microdata.
- Collaborate with mid-level and senior data engineers in Agile routines: daily stand-ups, sprint planning, and code reviews.

Projects

Economic Risk Monitor — GitHub
Data Engineering project — *Personal*

- Economic data pipeline built on the Medallion Architecture (Bronze/Silver/Gold), separating raw ingestion, cleaning, and the analytics layer.

Python Analytics Dashboards — GitHub
Streamlit, Pandas, Pyreadstat — *Personal*

- Interactive apps to process and explore surveys and complex datasets, including reading statistical files (SPSS/SAS) with Pyreadstat.

Skills

- **Programming:** Python, SQL
- **Data:** Pandas, Polars, DuckDB, SQLAlchemy, Pyreadstat, Playwright, REST APIs, YAML, Oracle, Data Quality, dbt, Medallion Architecture, Data Warehouse/Data Lake
- **Infrastructure & Tools:** Git/GitHub, Docker, Linux (Ubuntu/WSL), CLI, VS Code, Obsidian
- **Languages:** Portuguese (Native), English (Advanced)